A Short Guide to Attention

01 / Why attention?

Reading Tree original teaching material

A sequence model needs a way to combine information from different positions. A fixed-size summary can become a bottleneck: details that matter for one prediction may be irrelevant for another. Attention provides a flexible alternative. It computes a weighted combination of available information, with the weights depending on the current query.

In this guide we move from an intuitive retrieval example to scaled dot-product attention and multi-head attention. Our goal is to understand what queries, keys, and values do, and why attention weights should not automatically be interpreted as causal explanations.

Learning outcomes: explain the roles of Q, K, and V; trace the dimensions of the attention operation; describe the motivation for scaling; identify a limitation of interpreting attention maps.

02 / The idea of attention

Attention can be understood as content-based retrieval. A query represents what information is needed. Keys describe the available items, and values carry the information to be combined. Similarity scores between the query and each key are transformed into nonnegative weights that sum to one. The output is a weighted sum of the values.

Consider reading the sentence: The animal did not cross the street because it was tired. To build a useful representation for it, a model may draw on other words in the sentence. Attention allows every query position to combine information from other positions. This is a teaching example, not an empirical claim about a particular trained model.

A library analogy: the query is your question, keys are catalogue descriptions, and values are the actual book contents. In a neural network these are vectors, not literal questions or books. The analogy illustrates retrieval, but does not explain how the representations are learned.

02 / Queries, keys, and values

In self-attention, queries, keys, and values are derived from the same input sequence using different learned linear projections: Q = X W_Q, K = X W_K, V = X W_V. In cross-attention, queries and key-value pairs may come from different sequences.

Each attention output mixes value vectors. Similarity is computed using queries and keys. Keeping separate projections lets the model learn different spaces for matching and for passing information. The projections are learned jointly with the rest of the network.

Attention weights describe the mixing performed by this operation. They do not by themselves establish which tokens caused a final decision. Other heads, layers, residual connections, and nonlinear transformations also affect the result.

03 / Scaled dot-product attention

Attention(Q, K, V) = softmax(Q K^T / sqrt(d_k)) V

For n query positions and m key positions, Q has shape n by d_k, K has shape m by d_k, and V has shape m by d_v. Q K^T produces an n by m score matrix. A row-wise softmax converts each query row into weights. Multiplication by V produces an n by d_v output.

If independent query and key components have zero mean and unit variance, the variance of their dot product grows with d_k. Dividing by sqrt(d_k) keeps the scale more stable under these simplifying assumptions. This can help avoid an overly peaked softmax early in training.

A mask can remove disallowed positions before softmax. A causal mask prevents a position from attending to future positions. Padding masks prevent padded positions from contributing.

04 / Multiple perspectives

Multi-head attention performs several attention operations using separate learned projections. Each head can represent a different pattern of relationships, although there is no guarantee that a head corresponds to a clean human-interpretable concept.

The outputs of the heads are concatenated and projected again: MultiHead(Q,K,V) = Concat(head_1,...,head_h) W_O. The output projection combines information gathered by the different heads.

Self-attention alone does not encode sequence order. Positional information must be supplied through additional mechanisms. Attention also does not replace all computation in a Transformer: feed-forward blocks, normalization, and residual paths are important parts of the architecture.

05 / Check your understanding

1. Which vectors determine the attention scores, and which vectors are mixed to produce the output?
2. If Q is 8 by 64, K is 12 by 64, and V is 12 by 32, what are the shapes of the score matrix and the final output?
3. Why is the scaling factor sqrt(d_k) instead of d_k under the simplifying variance argument?
4. Why is an attention heatmap not a complete causal explanation of a model prediction?

Suggested answers: Scores use queries and keys; the output mixes values. The score matrix is 8 by 12 and the output is 8 by 32. The standard deviation, not the variance, scales with sqrt(d_k). Attention describes one mixing operation inside a larger computation, so causal claims require additional evidence.

This is an original educational example created for the Reading Tree workspace. It is not a reproduction or translation of a published paper.